The highest order derivative involved here is 2, and its power is 1 in the equation. Thus, the order of the differential equation is 2 and degree is 1.

(2) Consider the differential equation $\sqrt{1+\left(\frac{dy}{dx}\right)^2} = y\frac{d^3y}{dx^3}$.

Since this equation involves fractional powers, we must first get rid of them. On squaring the equation, we get

$$1+\left(\frac{dy}{dx}\right)^2 = y^2\left(\frac{d^3y}{dx^3}\right)^2 .$$

Now, we can clearly make out that the highest order derivative is 3. Therefore order of the differential equation is 3 and since its power is 2 in the equation, the degree of the differential equation is 2.

(3) Consider the differential equation $\sin\left(\frac{dy}{dx}\right)+\frac{d^2y}{dx^2}+3x=0$.

Here, the highest order derivative is 2. Because of sine of first derivative, the given differential equation can not be expressed as polynominal equation. So, the order of the differential equation is 2, and, it is not in polynomial equation in derivatives and so degree is not defined.

(4) Consider the equation $e^{\frac{d^2y}{dx^2}}+\sin(x)\frac{dy}{dx}=2$.

Here, the highest order derivative (order is 2) has involvement in an exponential function. This cannot be expressed as polynomial equation with $\frac{d^2y}{dx^2}$ as the leading term So, the degree of the equation is not defined. The order of the equation is 2.

(5) Further, the following differential equations do not have degrees.

(i) $e^{\frac{dy}{dx}}+\frac{dy}{dx}=0$ (ii) $\log\left(\frac{d^2y}{dx^2}\right)+\frac{dy}{dx}=0$ and (iii) $\cos\left(\frac{d^3y}{dx^3}\right)+2\frac{d^2y}{dx^2}=0$.

(6) The differential equation $10(y''')^4+7(y'')^5+\sin(y')+5=0$ has order 3 but degree is not defined.

(7) The differential equation $\cos(y')y'''+5y''+7y'=\sin x$ has order 3 and degree is not defined.

Remark

Observe that the degree of a differential equation is always a positive integer.

Example 10.1

Determine the order and degree (if exists) of the following differential equations:

(i) $\frac{dy}{dx}=x+y+5$ (ii) $\left(\frac{d^4y}{dx^4}\right)^3+4\left(\frac{dy}{dx}\right)^7+6y=5\cos 3x$

(iii) $\frac{d^2y}{dx^2}+3\left(\frac{dy}{dx}\right)^2=x^2\log\left(\frac{d^2y}{dx^2}\right)$ (iv) $3\left(\frac{d^2y}{dx^2}\right)=\left[4+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}$

(v) $dy+(xy-\cos x)dx=0$

Chapter 10 Differential Equation.indd 147 26-02-2020 17:23:31